

Mount Kenya University



Unlocking Infinite Possibilities

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DEPARTMENT OF INFORMATION TECHNOLOGY

UNIT CODE: BIT 3208

UNIT TITLE : ADVANCED WEB DESIGN

**PROJECT TITLE : CAMPUS EVENT BOARD : A CAMPUS EVENT
MANAGEMENT SYSTEM**

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Contents

1	WEEK 1: LOCAL DEVELOPMENT ENVIRONMENT SETUP	5
1.1	Installation of XAMPP	5
1.2	Starting Apache and MySQL Service	5
1.2.1	Reflection	6
2	WEEK 2 : WIREFRAMES, GUI DESIGN AND PROJECT PLANNING	7
2.1	Homepage Wireframe.....	7
2.2	Student Login Page Wireframe	8
2.3	Admin Dashboard Wireframe	8
2.4	Graphical User Interface Design	9
2.5	Project workflow for Campus Event Board.....	9
2.5.1	Technology stack used in the development of the system.....	10
		10
2.5.2	Reflection	11
3	WEEK 3: JavaScript and PHP Basics.....	12
3.1	Event Product Catalog	12
3.2	Form Validation.....	12
3.3	Password Strength Checker	13
3.4	DOM Manipulation	13
	Figure 10: DOM manipulation using JavaScript to interact with event content.	13
3.5	PHP Syntax Implementation.....	14
3.5.1	Reflection	14
4	WEEK 4 : SERVER-SIDE PROCESSING AND BACKEND DEVELOPMENT.....	15
4.1	Folder Organization and Project Structure	15
4.2	User Authentication Records	16
4.3	PHP Form Processing	17
4.4	Database Connection Configuration.....	18
4.5	GitHub Repository and Version Control.....	19
4.6	Weekly Reflection	19

5	WEEK 5: Database Integration and CRUD Operations	20
5.1	Database Creation	20
5.2	Database Tables	21
5.3	Create Operation	22
5.4	Read Operation.....	23
5.5	Update Operation.....	24
5.6	Delete Operation.....	24
5.7	SQL Query Execution.....	25
5.8	User Authentication Records	25
5.9	GitHub Repository Update	26
5.9.1	Reflection	26
6	WEEK 6: Advanced Event Management Features	27
6.1	Event Search Functionality.....	27
6.2	Featured and Upcoming Events Separation	28
6.3	Administrative Dashboard and Event Analytics	29
6.4	Image Upload for Events	29
6.5	Search Bug Debugging.....	30
6.5.1	Reflection	30
	Creative Enhancement — Featured Event Highlight.....	31
7	WEEK 7: Project Testing, Optimization and Deployment Preparation	32
7.1	Responsive Design Validation	32
7.2	End-to-End System Testing	33
7.3	User Interface and Navigation Validation.....	34
7.4	Final System Performance Validation.....	35
7.5	Final Project Verification	36
7.6	Creative Enhancement	37
7.6.1	Reflection	37
8	WEEK 8: Responsive Web Design and Mobile-First Development.....	38
8.1	Mobile-First CSS Optimization Using Media Queries	38

8.2	Responsive Layout Using Flexbox and CSS Grid.....	39
8.2.1	Reflection	39

1 WEEK 1: LOCAL DEVELOPMENT ENVIRONMENT SETUP

1.1 Installation of XAMPP

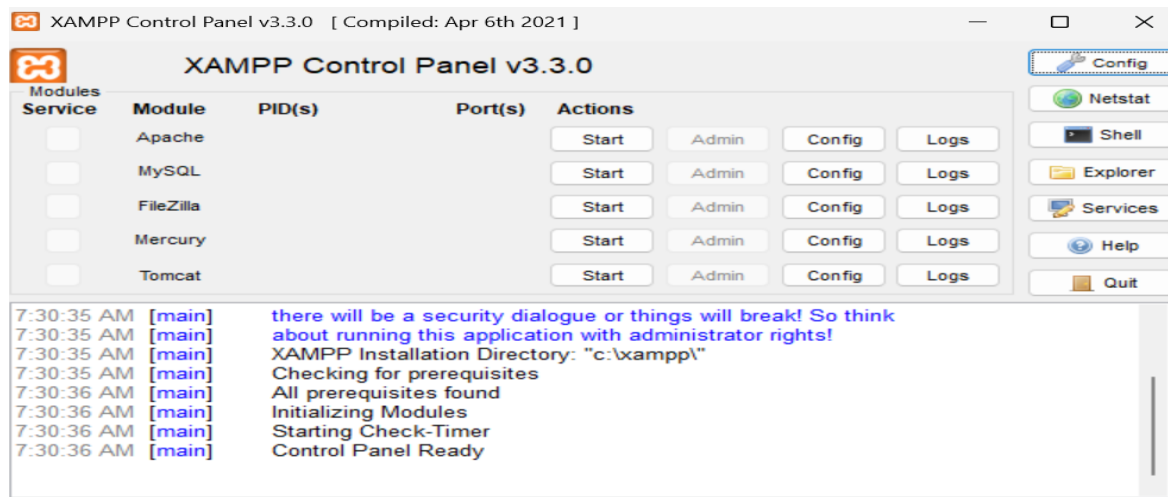


Figure 1: installation of XAMPP

1.2 Starting Apache and MySQL Service

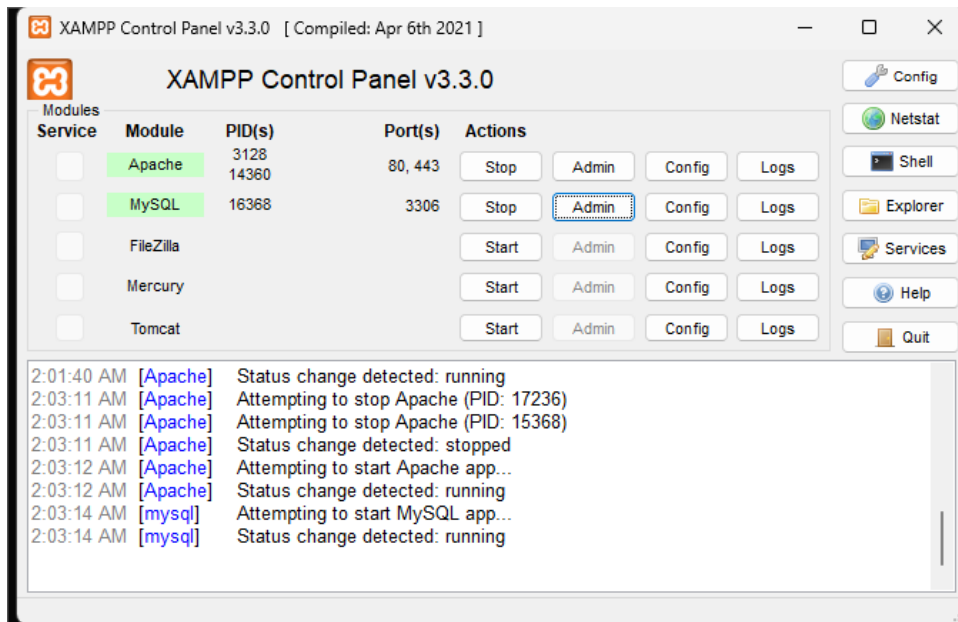


Figure 2: Apache and MySQL running

1.2.1 Reflection

During Week 1, I successfully set up XAMPP and configured Apache and MySQL for local web development. I learned how localhost works and how PHP applications communicate with a MySQL database. The main challenge was understanding project deployment within the htdocs directory.

2 WEEK 2: WIREFRAMES, GUI DESIGN AND PROJECT PLANNING

2.1 Homepage Wireframe

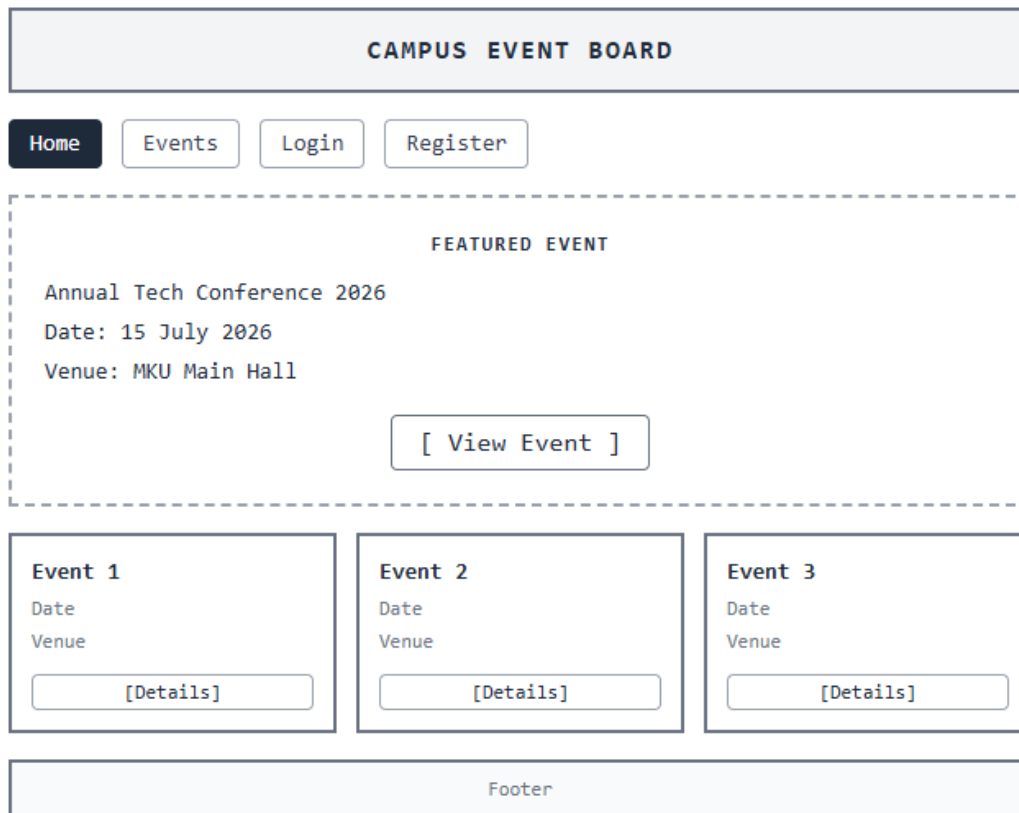


Figure 3:Homepage Wireframe for the Campus Event Board system.

2.2 Student Login Page Wireframe

CAMPUS EVENT BOARD

Student Login

Email

Password

[Login]

Don't have an account?

[Register]

Footer

Figure 4: Student Login Page Wireframe.

2.3 Admin Dashboard Wireframe

ADMIN DASHBOARD

Dashboard Events RSVPs Logout

12
Total Events

348
Total Students

521
Total RSVPs

Event Name	Date	Venue	Edit	Delete
Tech Day	...	...	[E]	[D]
Career Day	...	...	[E]	[D]

[Create New Event]

Figure 5: Admin Dashboard Wireframe.

2.4 Graphical User Interface Design

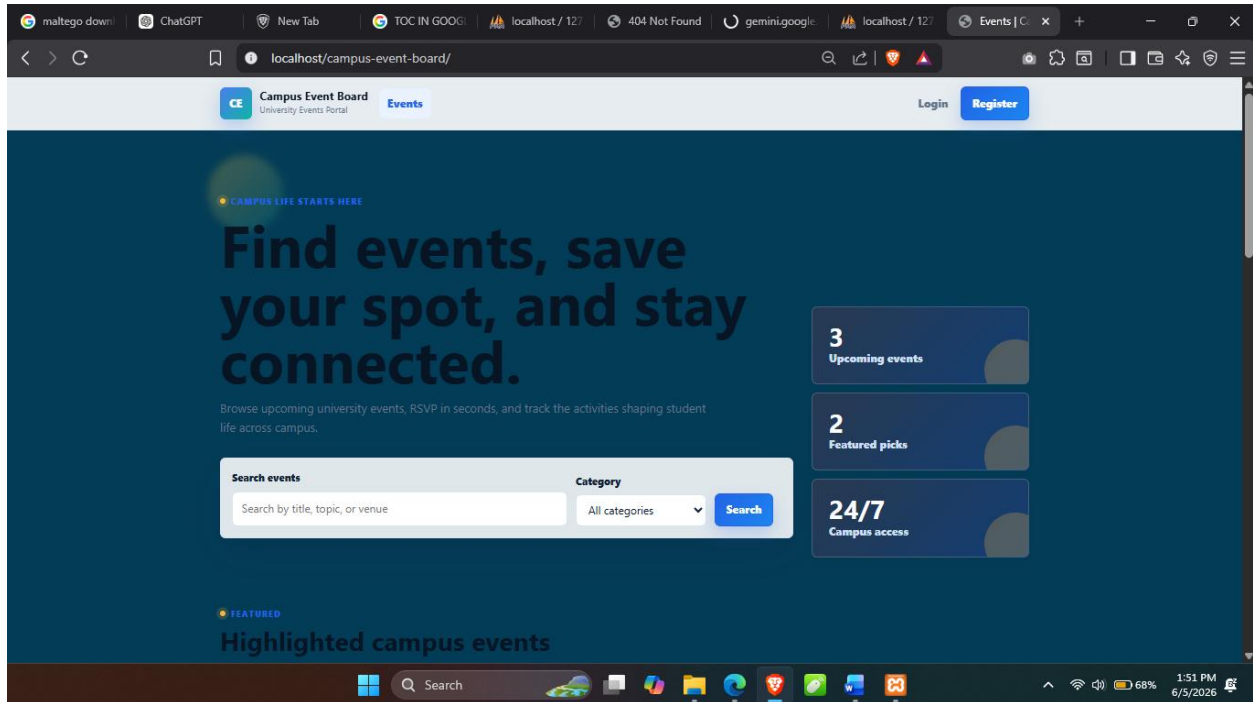
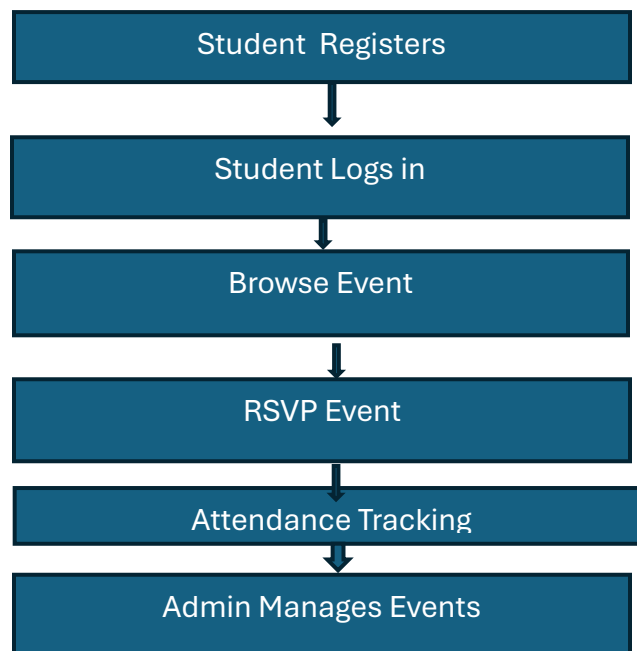
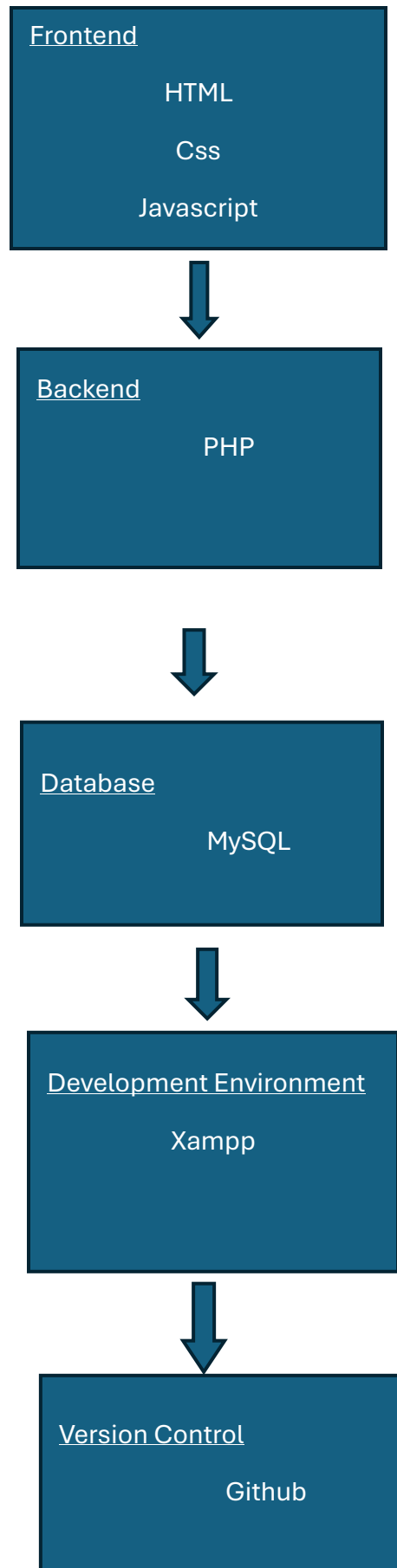


Figure 6: shows the Graphics User Interface of the campus event management system .

2.5 Project workflow for Campus Event Board.



2.5.1 Technology stack used in the development of the system



2.5.2 Reflection

During Week 2, I designed wireframes, GUI layouts, and a workflow for the Campus Event Board system. This helped me visualize system functionality before development. I also identified the technologies required for implementation, including HTML, CSS, JavaScript, PHP, and MySQL.

3 WEEK 3: JavaScript and PHP Basics

3.1 Event Product Catalog

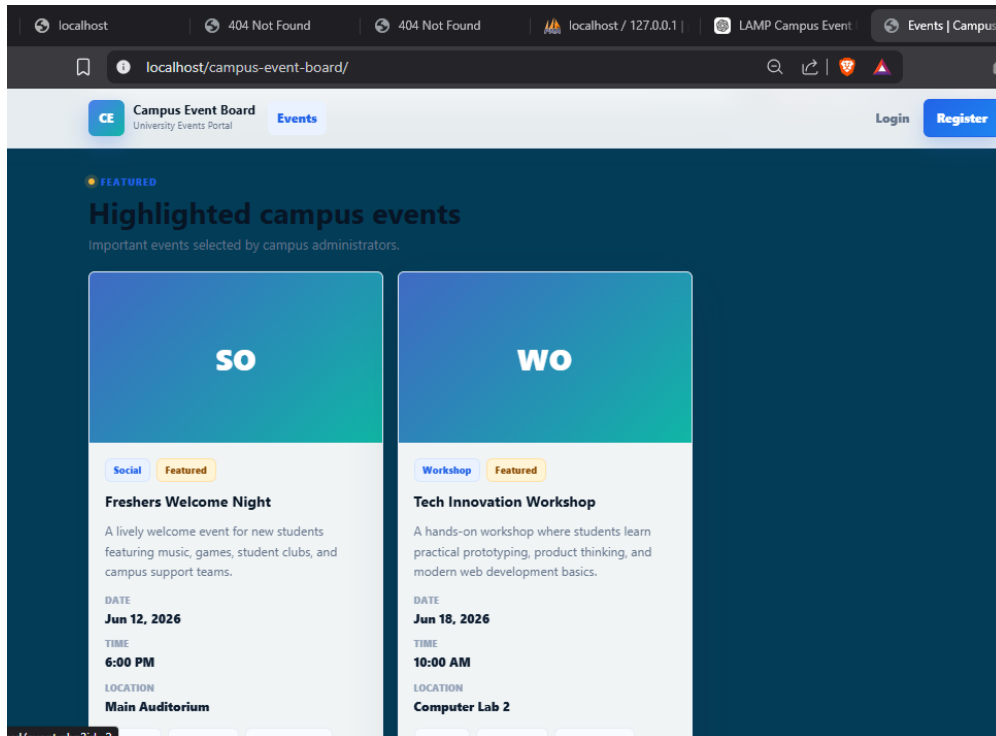


Figure 7: Event catalog developed using HTML, CSS and JavaScript.

3.2 Form Validation

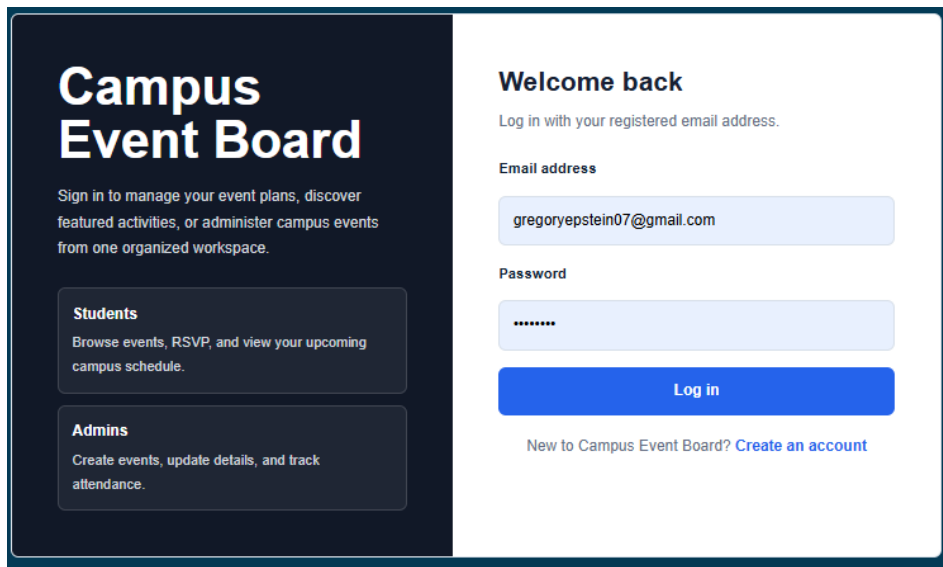


Figure 8: Login form validation implementation.

3.3 Password Strength Checker

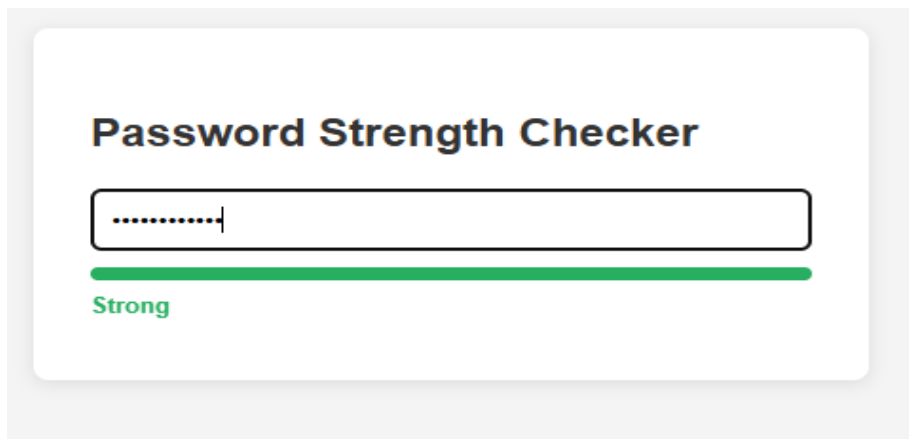


Figure 9: Password Strength Checker implementation

3.4 DOM Manipulation

```
SourceCode > assets > js > JS app.js > ...
1  document.addEventListener('DOMContentLoaded', () => {
2      initMobileNavigation();
3      initLiveEventSearch();
4      initConfirmations();
5      initImagePreview();
6      initFormValidation();
7      initModals();
8  });
9
10 /* Full JavaScript saved in your project file */
```

Figure 10: DOM manipulation using JavaScript to interact with event content.

3.5 PHP Syntax Implementation

```
SourceCode > event.php > section.content-section.event-info-section
1  <?php
2  declare(strict_types=1);
3
4  require_once __DIR__ . '/config/db.php';
5  require_once __DIR__ . '/includes/auth.php';
6
7  $eventId = filter_input(INPUT_GET, 'id', FILTER_VALIDATE_INT);
8
9  if (!$eventId) {
10     header('Location: index.php');
11     exit;
12 }
13
14 $eventStatement = $pdo->prepare(
15     'SELECT
16         e.*,
17         COUNT(DISTINCT CASE WHEN r.status = "going" THEN r.id END) AS rsvp_count,
18         COUNT(DISTINCT a.id) AS attendance_count
19     FROM events e
20     LEFT JOIN rsmps r ON r.event_id = e.id
21     LEFT JOIN attendance a ON a.rsvp_id = r.id
22     WHERE e.id = :event_id
23     GROUP BY e.id
24     LIMIT 1'
25 );
26 $eventStatement->execute([
27     ':event_id' => $eventId,
28 ]);
29 $event = $eventStatement->fetch();
30
31 if (!$event) {
32     header('Location: index.php');
```

Figure 11: PHP syntax implementation for dynamic content generation.

3.5.1 Reflection

During Week 3, I implemented JavaScript and PHP features including an event catalog, form validation, password strength checking, and DOM manipulation. I gained practical experience in creating interactive web pages and using PHP to generate dynamic content for web applications.

4 WEEK 4: SERVER-SIDE PROCESSING AND BACKEND DEVELOPMENT

4.1 Folder Organization and Project Structure

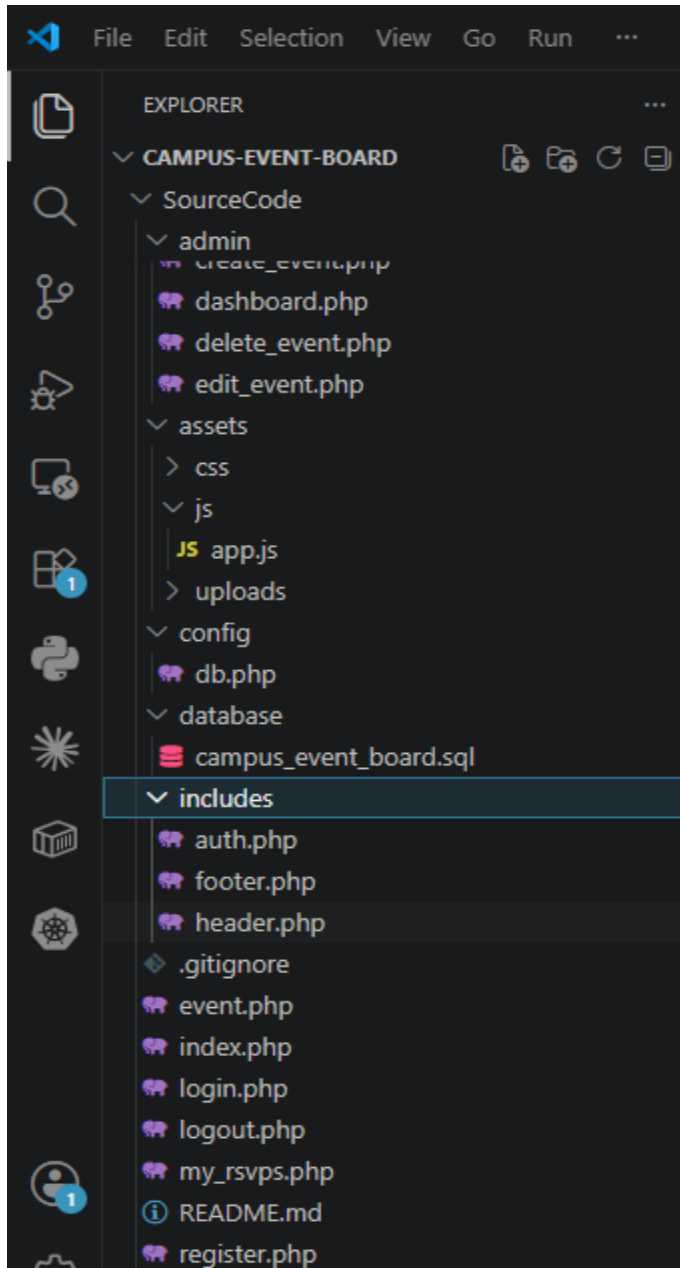
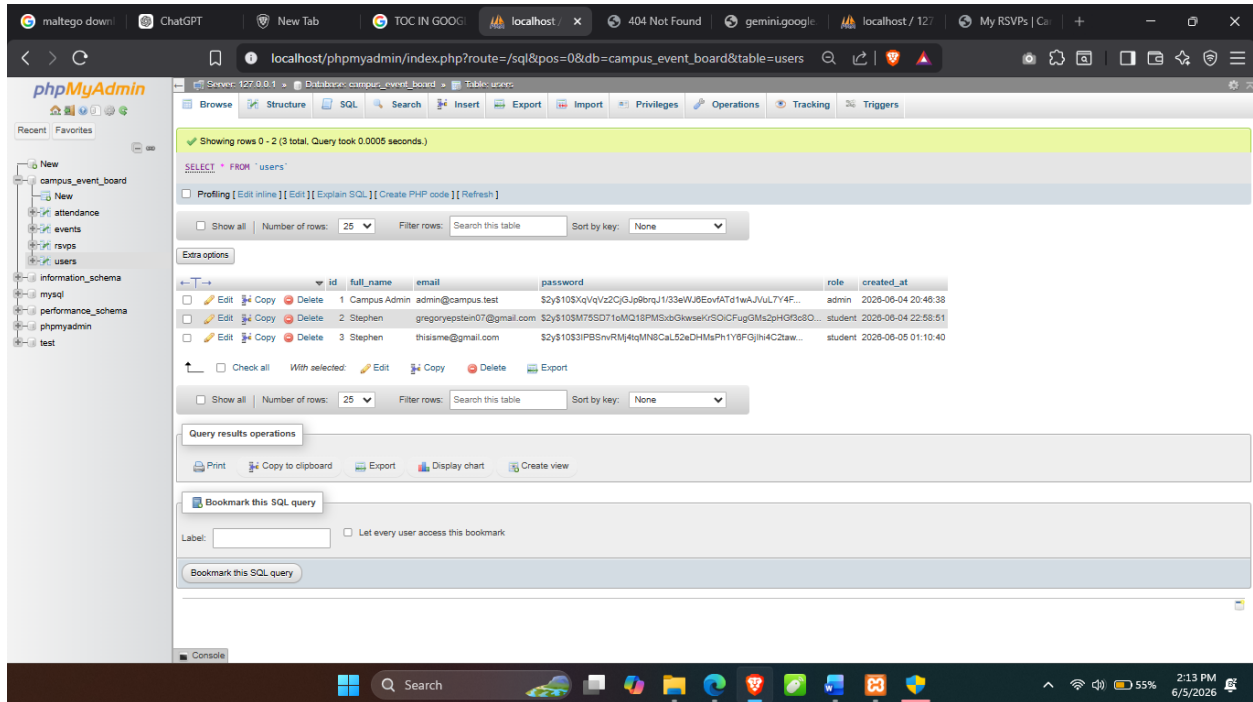


Figure 12: Folder organization and project structure for the Campus Event Board system.

4.2 User Authentication Records



The screenshot displays the phpMyAdmin interface for a MySQL database named 'campus_event_board'. The 'users' table is selected, and the query results show three rows of user authentication records. The table structure includes columns for id, full_name, email, password, role, and created_at. The records are as follows:

id	full_name	email	password	role	created_at
1	Campus Admin	admin@campus.test	\$2y\$10\$KqVqV2CjGj9BrqJ1/33eWJ9EovfATd1eAJuLTy4F...	admin	2020-06-04 20:48:38
2	Stephen	gregoryepsten07@gmail.com	\$2y\$10\$M75SD71oMQ18PM5ibGkweK/SOICFugOMs2pHOF3c8O...	student	2020-06-04 22:58:51
3	Stephen	thisisme@gmail.com	\$2y\$10\$3IPB5nRMj4tqMNBcaL52eCHMsPhY19F0j/h4C2aw...	student	2020-06-05 01:10:40

The interface also shows the SQL query used to retrieve the data: `SELECT * FROM 'users'`. Below the table, there are options for query results operations (Print, Copy to clipboard, Export, Display chart, Create view) and a bookmark section.

Figure 13: User authentication records stored in the MySQL database.

4.3 PHP Form Processing

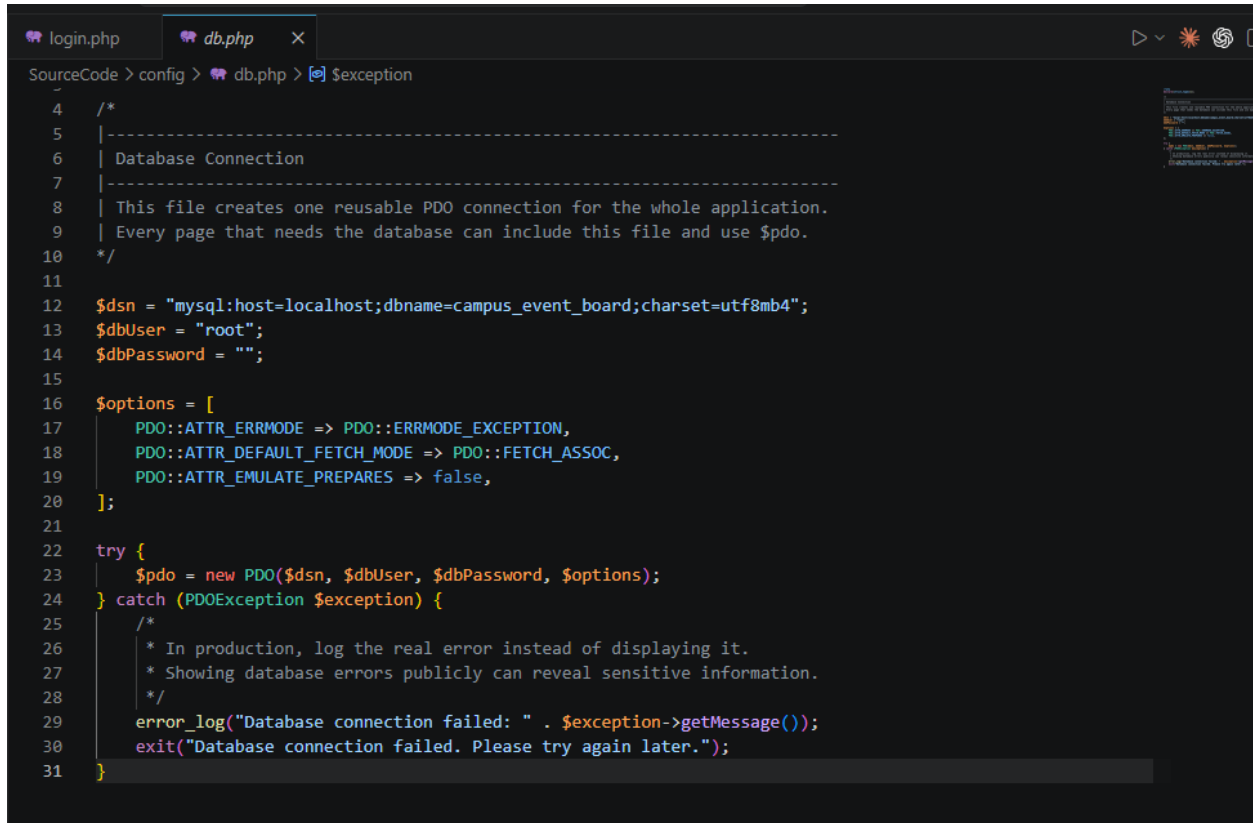
```
SourceCode > login.php > html
12
13     header('Location: index.php');
14     exit;
15 }
16
17 $email = '';
18 $errors = [];
19
20 if ($_SERVER['REQUEST_METHOD'] === 'POST') {
21     $email = trim($_POST['email'] ?? '');
22     $password = $_POST['password'] ?? '';
23
24     /*
25      * Keep login errors general. This avoids telling attackers whether an
26      * email address exists in the system.
27      */
28     $loginError = 'Invalid email or password.';
29
30     if ($email === '' || $password === '') {
31         $errors[] = 'Email and password are required.';
32     } elseif (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
33         $errors[] = $loginError;
34     } else {
35         $userStatement = $pdo->prepare(
36             'SELECT id, full_name, email, password, role
37             FROM users
38             WHERE email = :email
39             LIMIT 1'
40         );
41         $userStatement->execute([
42             ':email' => $email,
43         ]);
44     }
45 }
```

Stephen Kimani (6 days ago) Ln 357, Col 8 Spaces: 4 UTF-8 CRLF PHP Go Live Prettier

Search 1:45 PM 6/19/2026

Figure 14: PHP processing of user login requests using POST data.

4.4 Database Connection Configuration



```
login.php db.php x
SourceCode > config > db.php > $exception
4  /*
5  |-----
6  | Database Connection
7  |-----
8  | This file creates one reusable PDO connection for the whole application.
9  | Every page that needs the database can include this file and use $pdo.
10 */
11
12 $dsn = "mysql:host=localhost;dbname=campus_event_board;charset=utf8mb4";
13 $dbUser = "root";
14 $dbPassword = "";
15
16 $options = [
17     PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
18     PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
19     PDO::ATTR_EMULATE_PREPARES => false,
20 ];
21
22 try {
23     $pdo = new PDO($dsn, $dbUser, $dbPassword, $options);
24 } catch (PDOException $exception) {
25     /*
26     * In production, log the real error instead of displaying it.
27     * Showing database errors publicly can reveal sensitive information.
28     */
29     error_log("Database connection failed: " . $exception->getMessage());
30     exit("Database connection failed. Please try again later.");
31 }
```

Figure 15: Database connection configuration using PDO.

4.5 GitHub Repository and Version Control

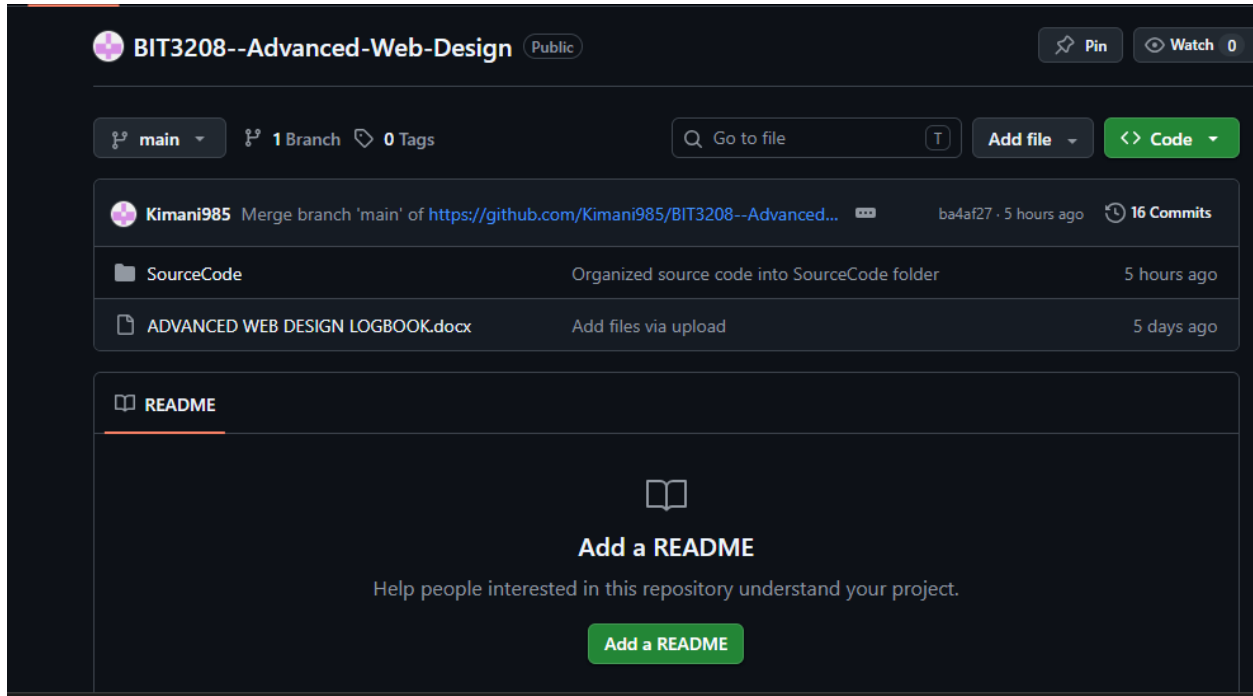


Figure 16: GitHub repository showing project version control.

4.6 Weekly Reflection

During Week 4, I learned how server-side processing works by integrating PHP with HTML forms and a MySQL database. I implemented login functionality, database connectivity using PDO, and user authentication mechanisms. I also organized the project structure professionally and used GitHub for version control and project backup.

5 WEEK 5: Database Integration and CRUD Operations

5.1 Database Creation

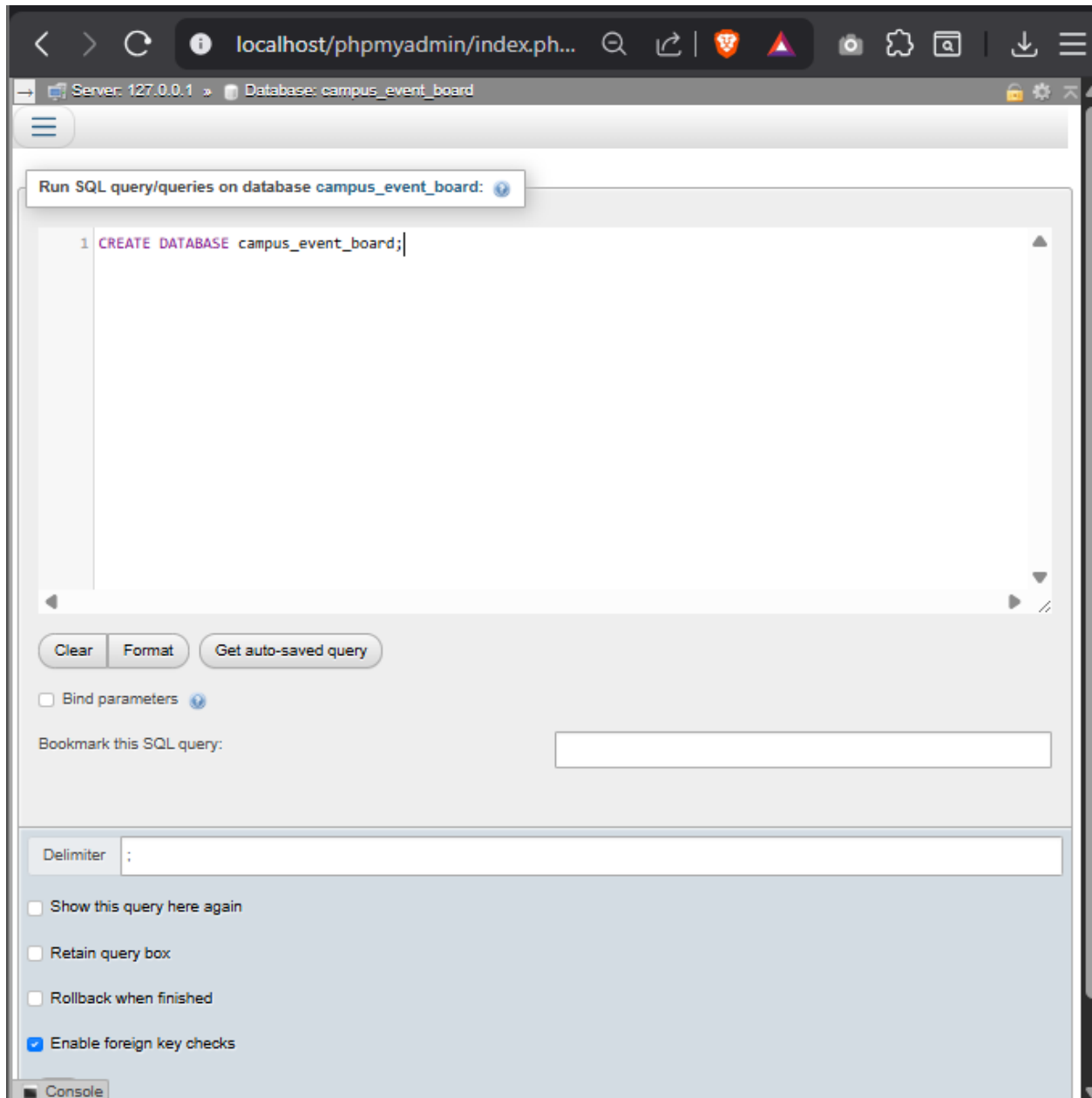
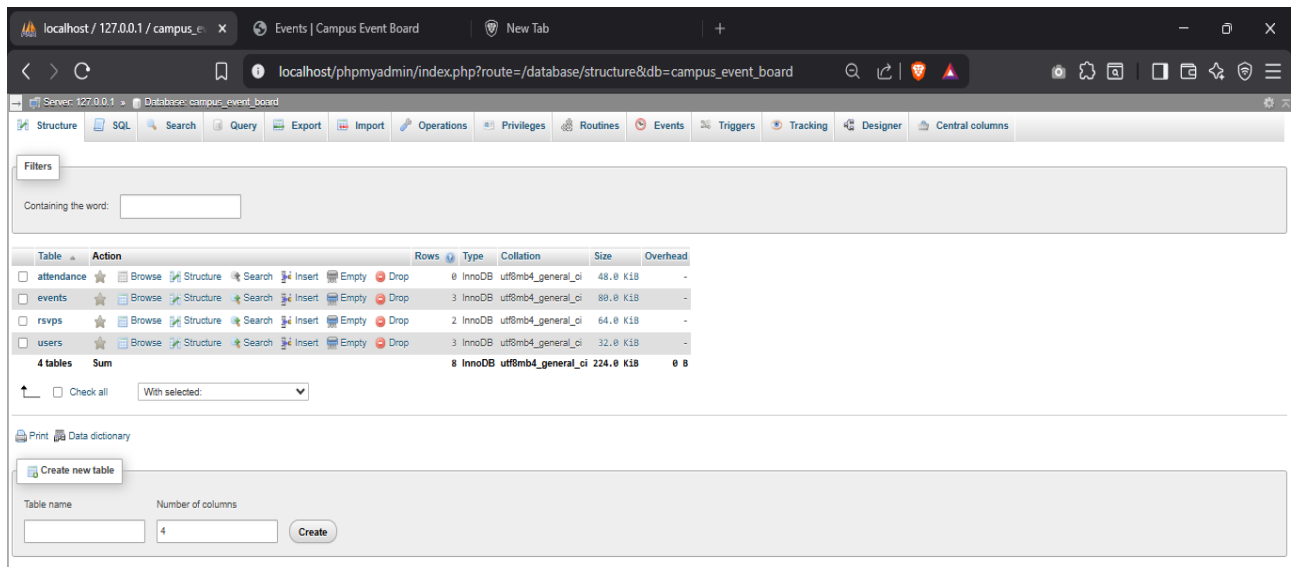


Figure 17: Creation of the Campus Event Board database using phpMyAdmin.

5.2 Database Tables



The screenshot shows the phpMyAdmin interface for a database named 'campus_event_board'. The 'Structure' tab is selected, displaying a table of database tables. The table has columns: Table, Action, Rows, Type, Collation, Size, and Overhead. The tables listed are 'attendance', 'events', 'rvsps', and 'users'. Below the table, there is a '4 tables' summary row. At the bottom, there is a 'Create new table' form with fields for 'Table name' and 'Number of columns' (set to 4), and a 'Create' button.

Table	Action	Rows	Type	Collation	Size	Overhead
☐ attendance	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	48.0 K1B	-
☐ events	Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_general_ci	88.0 K1B	-
☐ rvsps	Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_general_ci	64.0 K1B	-
☐ users	Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_general_ci	32.0 K1B	-
4 tables Sum		8	InnoDB	utf8mb4_general_ci	224.0 K1B	0 B

☐ Check all With selected: ▼

[Print](#) [Data dictionary](#)

[Create new table](#)

Table name: Number of columns: [Create](#)

Figure 18: Database tables created for the Campus Event Board system.

5.3 Create Operation

The screenshot shows a web browser at the URL `localhost/campus-event-board/SourceCode/admin/create_event.php`. The page has a dark blue header with the 'Campus Event Board' logo and navigation links for 'Events', 'Dashboard', and 'Create Event'. A 'Campus Admin' button with a dropdown arrow and a 'Logout' button are also present. The main content area is titled 'Create event' with a 'Back to dashboard' button. Below the title is a subtitle: 'Add a new campus event with details, capacity, image, and featured placement.' The form itself is divided into several sections: 'Event title' with a text input containing 'Team1 Africa Community call 16 🚩 Road to Mini Hack'; 'Description' with a text area containing three paragraphs about 'Team1 Africa Community Calls'; 'Category' with a dropdown menu set to 'TECH'; 'Venue' with a text input set to 'EVEREST HALL'; 'Date' with a date picker set to '07/18/2026'; 'Time' with a time picker set to '09:00 AM'; 'Capacity' with a text input set to '1000'; and 'Event image' with a 'Choose File' button and the filename 'ETHICAL HACKING COINFRENCE.png'. A note at the bottom right of the form states 'Optional .JPG, .PNG, or WEBP. Maximum 2MB.'

Figure 19: Create operation used to add a new event to the Campus Event Board system.

5.4 Read Operation

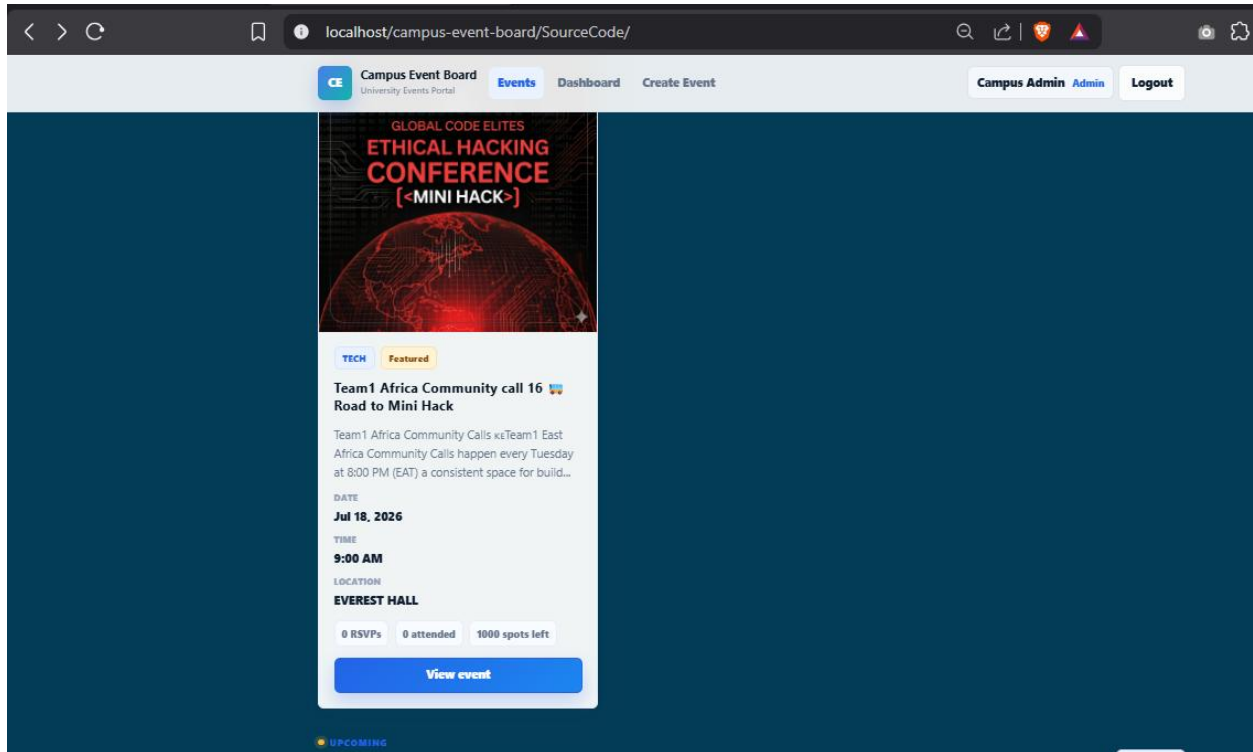


Figure 20: Read operation displaying events stored in the database.

5.5 Update Operation

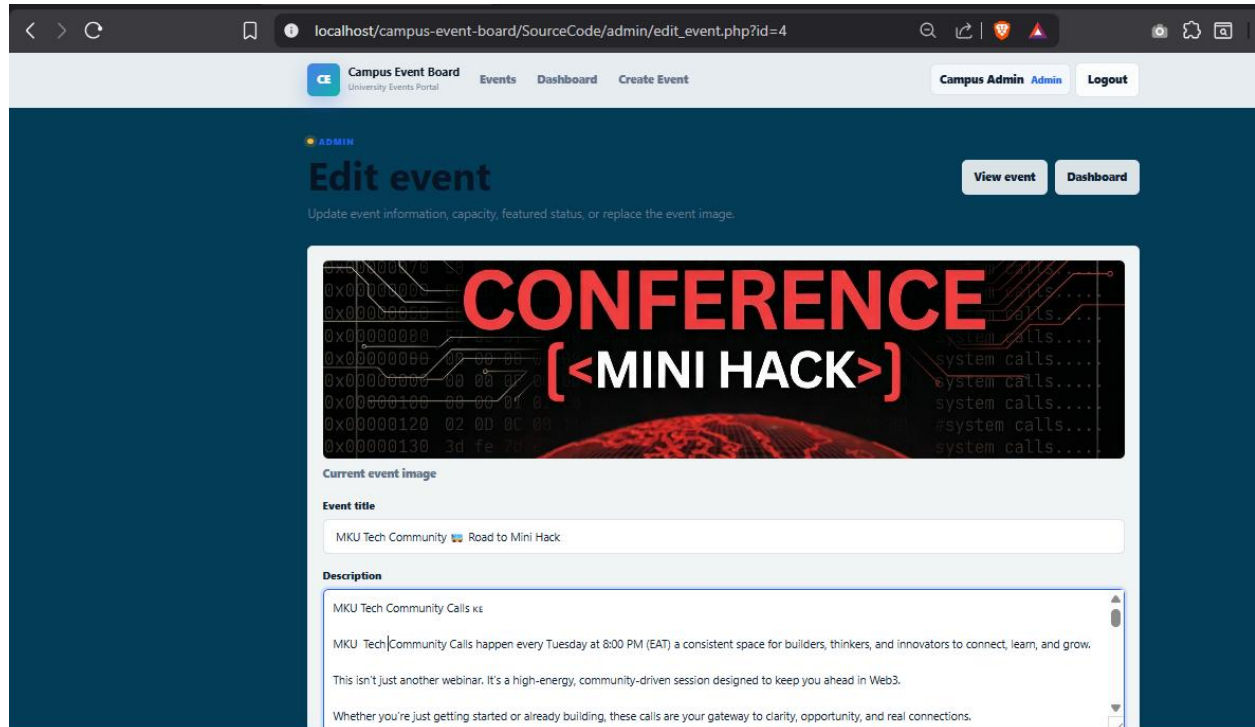


Figure 21: Update operation used to modify event information.

5.6 Delete Operation

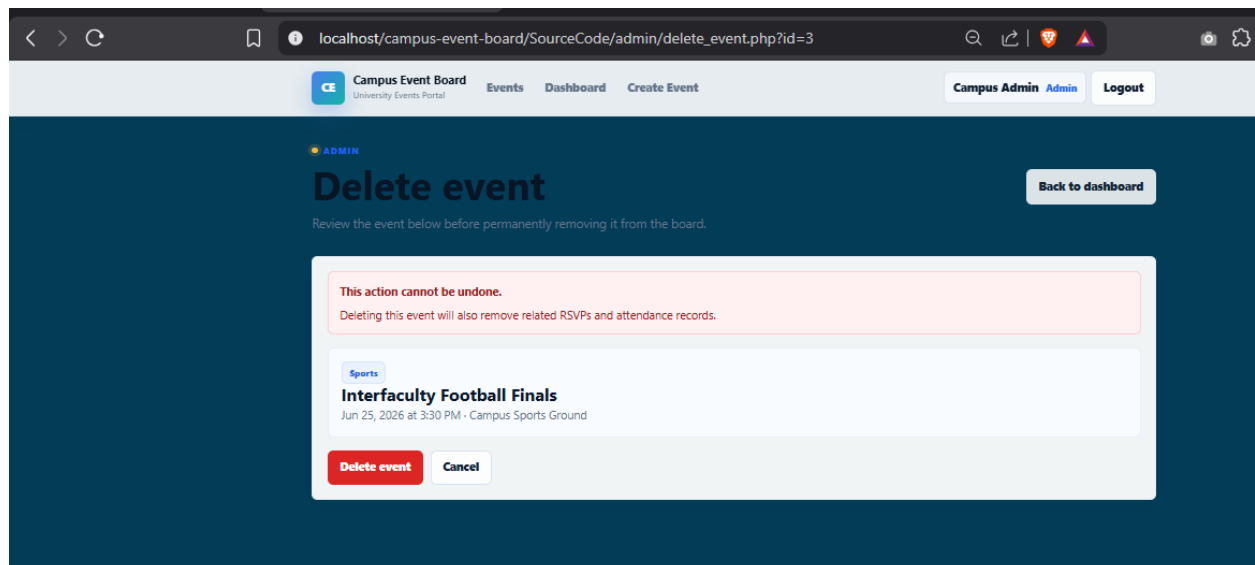


Figure 22: Delete operation available for removing event records from the system.

5.7 SQL Query Execution

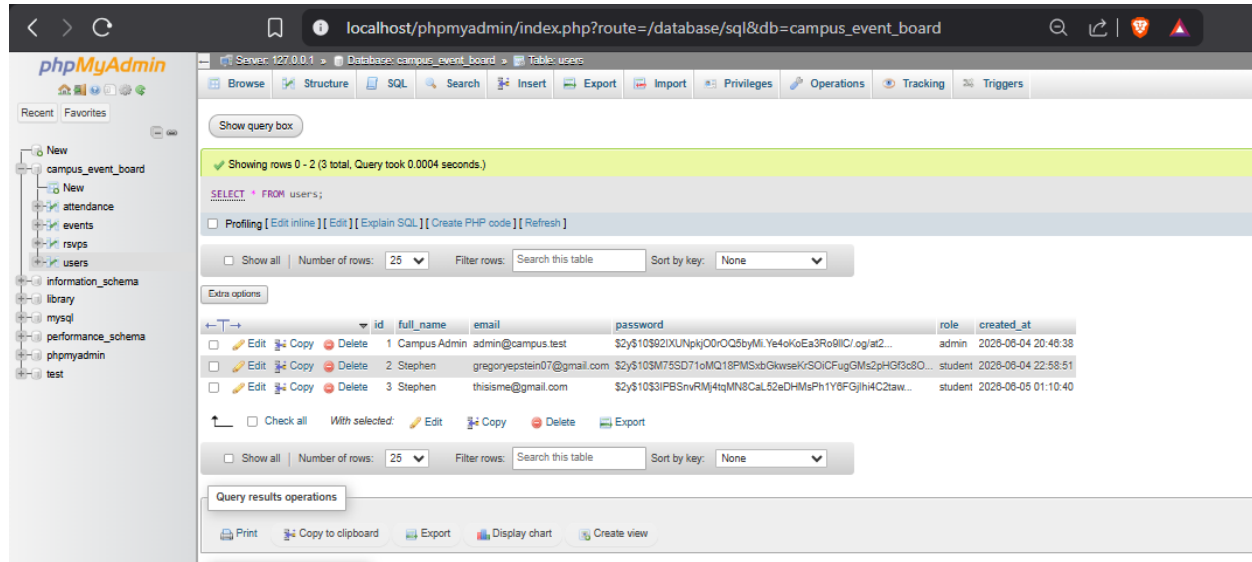


Figure 23: SQL query execution within phpMyAdmin.

5.8 User Authentication Records

Extra options						
	id	full_name	email	password	role	created_at
☐	1	Campus Admin	admin@campus.test	\$2y\$10\$92IXUNpkjO0rQ5byMi.Ye4oKoEa3Ro9llC/.og/at2...	admin	2026-06-04 20:46:38
☐	2	Stephen	gregoryepstein07@gmail.com	\$2y\$10\$M75SD71oMQ18PMSxbGkwseKrSOiCFugGMs2pHGf3c8O...	student	2026-06-04 22:58:51
☐	3	Stephen	thisisme@gmail.com	\$2y\$10\$3IPBSnvRMj4tqMN8CaL52eDHMsPh1Y8FGjih4C2taw...	student	2026-06-05 01:10:40

Figure 24: User authentication records stored in the Campus Event Board database.

5.9 GitHub Repository Update

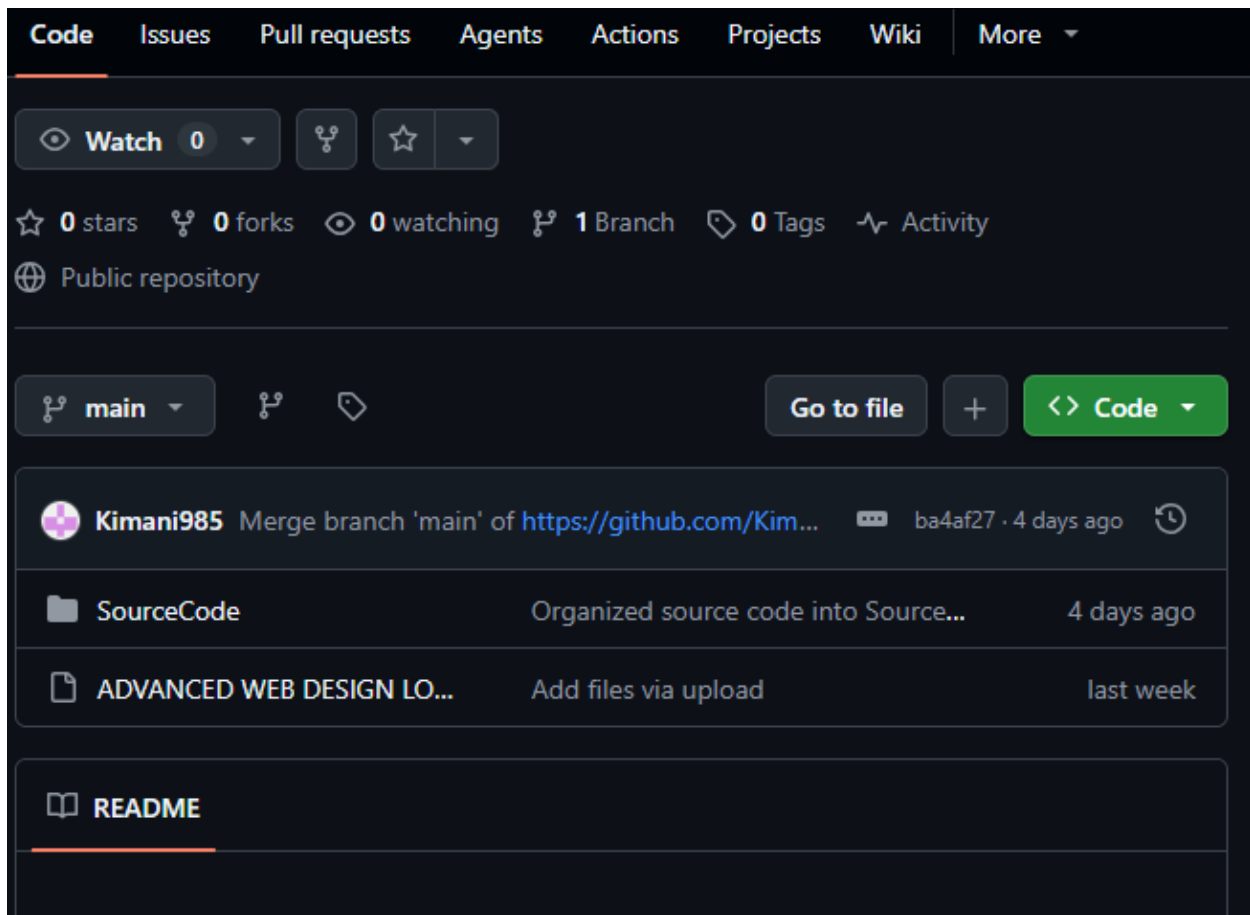


Figure 25: GitHub repository showing project source code, documentation, logbook, and creative implementation files.

5.9.1 Reflection

During Week 5, I integrated the Campus Event Board system with a MySQL database using PDO. CRUD operations were implemented and tested, enabling event creation, retrieval, modification, and deletion. SQL queries were executed through phpMyAdmin, and GitHub was used to manage and update project files.

6 WEEK 6: Advanced Event Management Features

6.1 Event Search Functionality

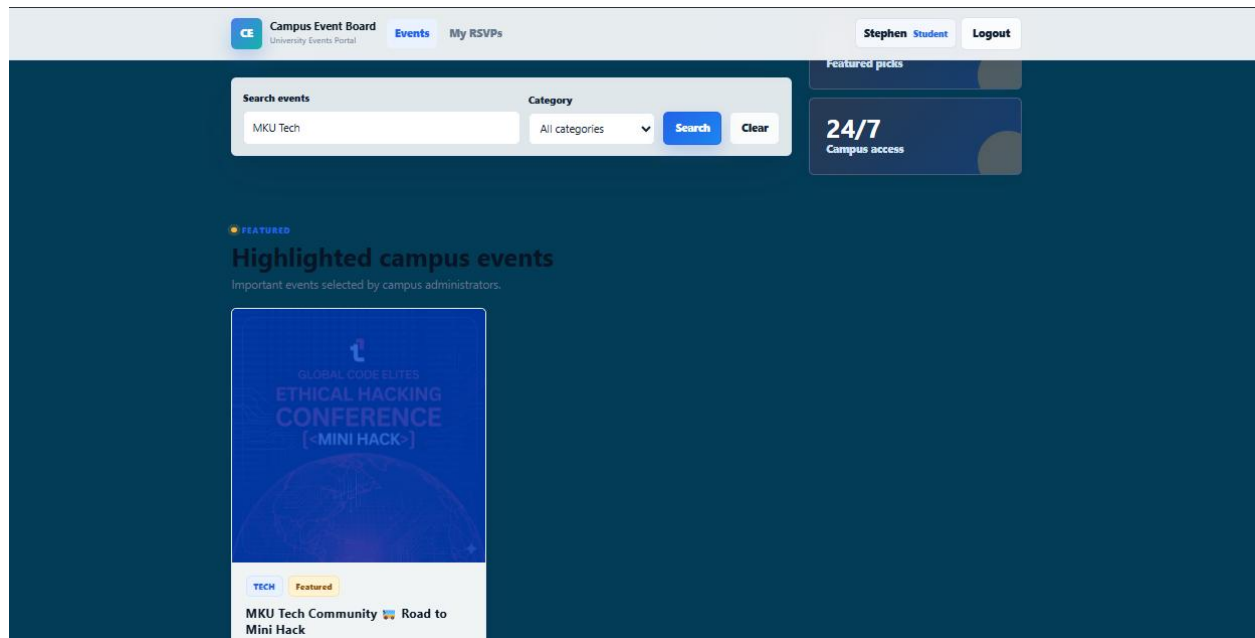


Figure 26: Event search and category filtering interface.

6.2 Featured and Upcoming Events Separation

The screenshot displays the 'Campus Event Board' interface. At the top, there's a navigation bar with 'Events' and 'My RSVPs' tabs, and a user profile 'Stephen Student' with a 'Logout' button. The main header features a large blue banner with the text 'Find events, save your spot, and stay connected.' Below this is a search bar with a placeholder 'Search by title, topic, or venue' and a 'Search' button. To the right of the search bar are three statistics: '5 Upcoming events', '3 Featured picks', and '24/7 Campus alerts'.

The 'Highlighted campus events' section follows, with a sub-header 'Important events selected by campus administrators.' It contains three event cards:

- GLOBAL CODE ELITES ETHICAL HACKING CONFERENCE (~MINI HACK~)**: A 'Tech' featured event on Jul 18, 2026, from 9:00 AM to 11:00 AM at EVEREST HALL. It has 2 RSVPs, 0 attended, and 28 spots left.
- Entrepreneurship & Startup Summit**: A 'Business' featured event on Sep 27, 2026, from 11:00 AM to 12:00 PM at Conference Hall A. It has 2 RSVPs, 0 attended, and 18 spots left.
- Wellness & Mental Health Awareness Day**: A 'Health' featured event on Sep 29, 2026, from 10:00 AM to 11:00 AM at Student Wellness Centre. It has 1 RSVP, 0 attended, and 99 spots left.

Below this is the 'All upcoming events' section, with a sub-header 'Search, filter, and choose what to keep on your campus calendar.' It shows a grid of events, including the same three featured events and two new ones:

- Cultural Heritage Festival**: A 'Culture' event on Oct 9, 2026, from 2:00 PM to 5:00 PM at Student Centre Grounds. It has 2 RSVPs, 0 attended, and 88 spots left.
- Talent & Music Showcase**: A 'Recreation' featured event on Oct 18, 2026, from 6:00 PM to 8:00 PM at University Amphitheatre. It has 1 RSVP, 0 attended, and 49 spots left.

The footer contains a 'Campus Event Board' logo, a description 'A modern university portal for discovering events, managing RSVPs, and tracking attendance.', and links to 'Explore' (Events, My RSVPs, Student Login) and 'Connect' (Facebook, Instagram, LinkedIn, YouTube). Copyright information '© 2026 Campus Event Board. All rights reserved.' and 'Built for secure campus event management.' are also present.

Figure 27:Featured and upcoming events displayed separately.

6.3 Administrative Dashboard and Event Analytics

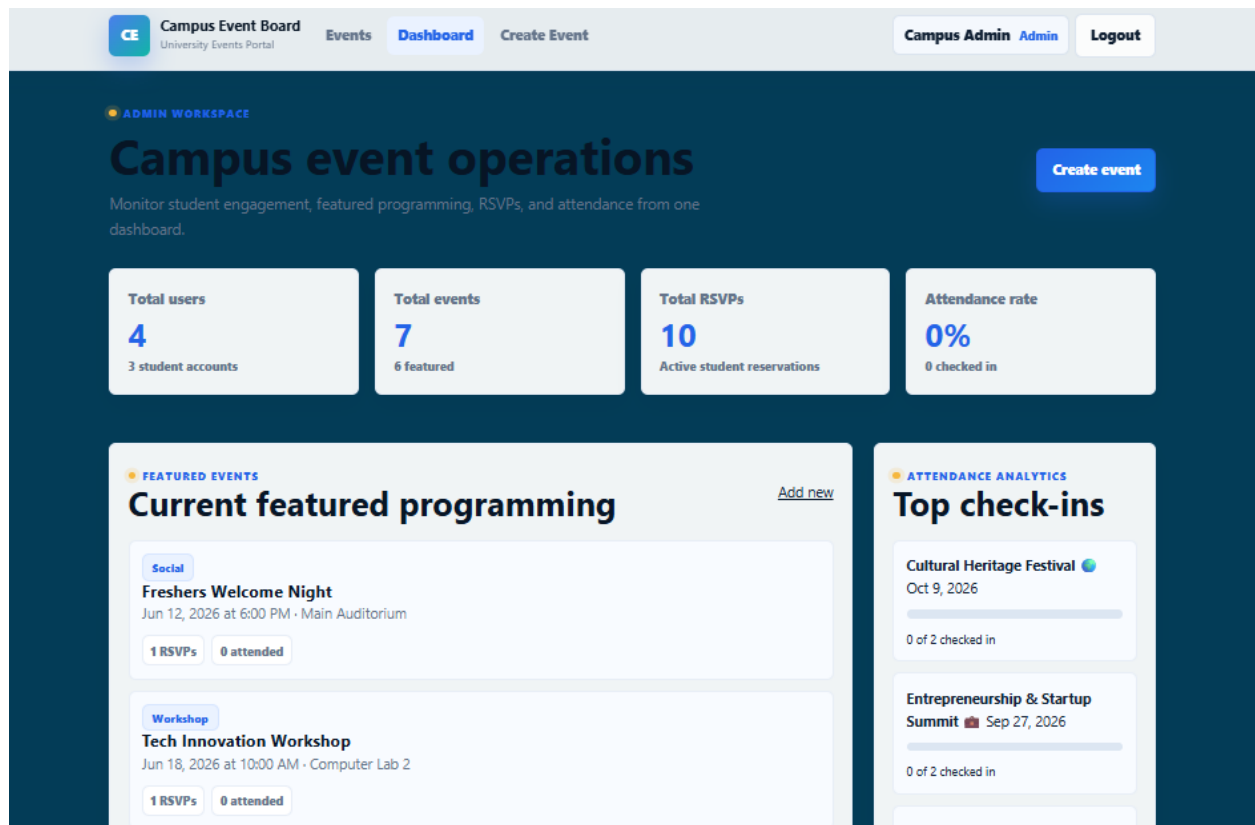


Figure 28: Administrative dashboard displaying event statistics, RSVP analytics, attendance information, and featured event management.

6.4 Image Upload for Events

The screenshot shows the 'Event image' section of the event creation form. It includes a 'Capacity' input field, an 'Event image' section with a 'Choose File' button and 'No file chosen' text, and a checkbox for 'Feature this event on the homepage'. Below these are 'Create event' and 'Cancel' buttons. A note specifies 'Optional. JPG, PNG, or WEBP. Maximum 2MB.'

Figure 29: Image upload feature during event creation.

6.5 Search Bug Debugging

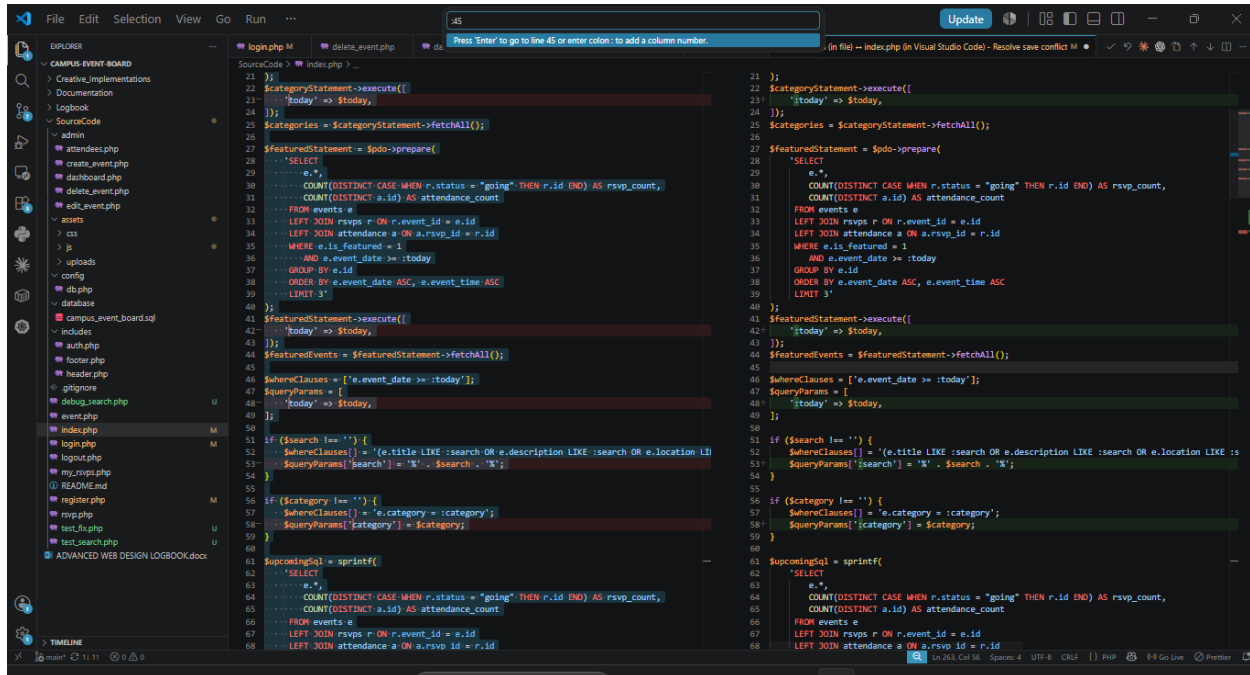


Figure 30: Debugging the event search functionality.

6.5.1 Reflection

During Week 6, I implemented advanced event management features including event search, category filtering, featured event separation, image upload, and debugging of SQL queries. These activities improved my understanding of dynamic content management, usability, and troubleshooting database-driven web applications.

Creative Enhancement — Featured Event Highlight

I enhanced the homepage by giving featured events special badges and making them appear before regular events to improve visibility.

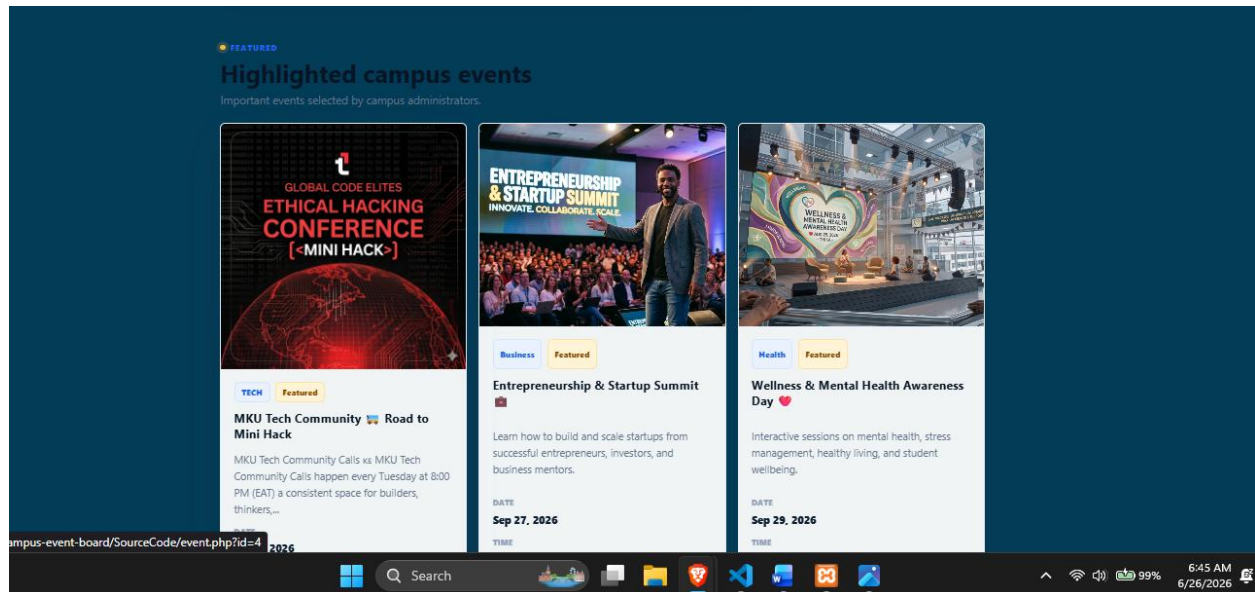


Figure 31: Featured events highlighted for better visibility.

7 WEEK 7: Project Testing, Optimization and Deployment Preparation

7.1 Responsive Design Validation

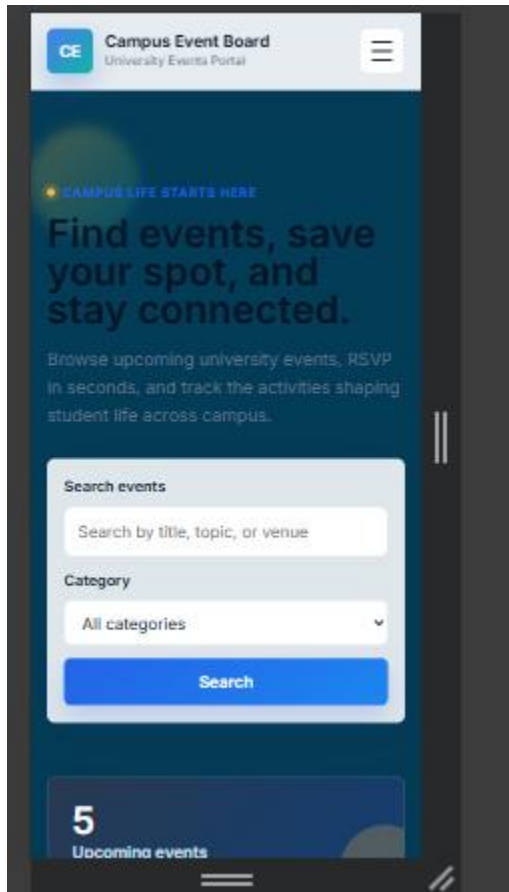


Figure 32: Responsive design testing using Chrome Developer Tools to validate compatibility across multiple screen sizes.

I tested the responsiveness of the Campus Event Board across different screen sizes to ensure users could access all features on desktop, tablet, and mobile devices. For example in that screenshot that's the iPhone 12 screen (390 × 844). This improved my understanding of responsive web design and user experience.

7.2 End-to-End System Testing

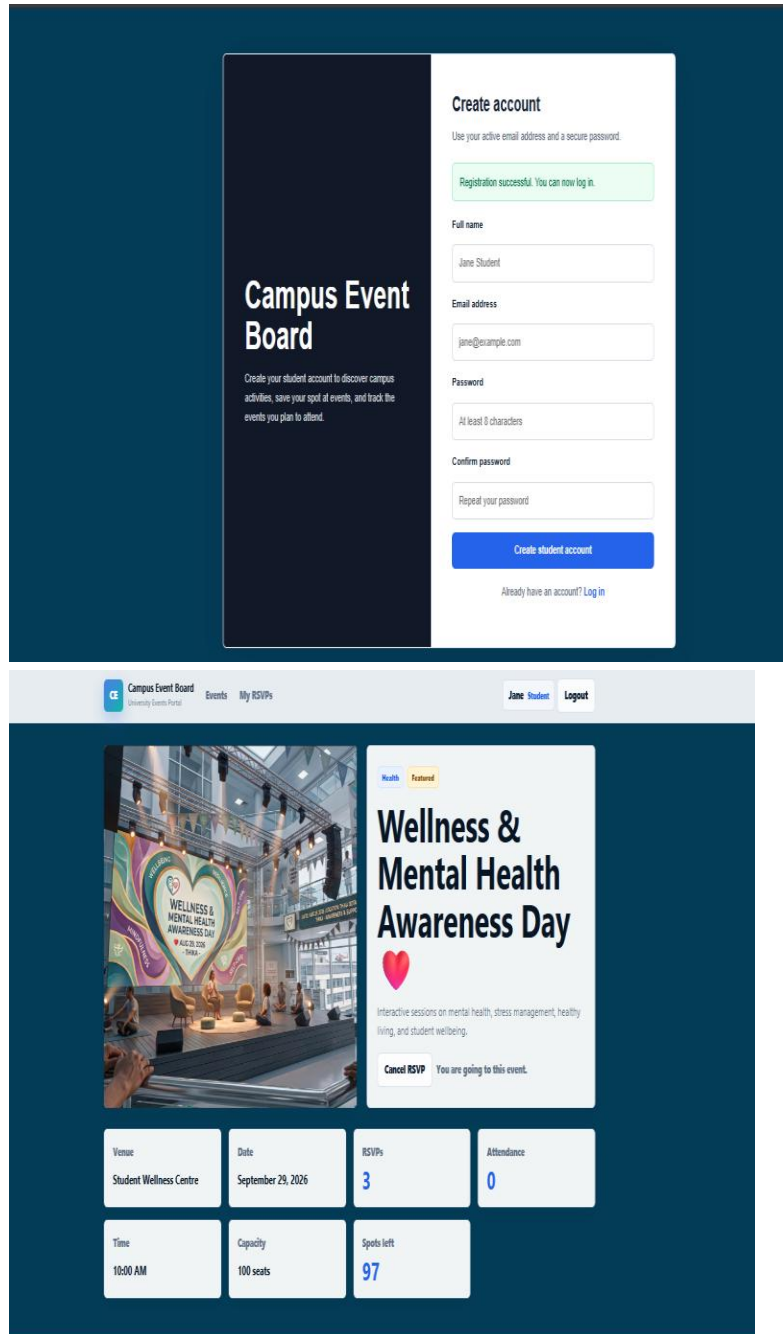


Figure 33: Successful end-to-end testing of the Campus Event Board, confirming that user registration, authentication, event management, and RSVP functionality work together as an integrated system

End-to-end testing was performed to ensure that all major components of the application interacted correctly. The testing confirmed successful user registration, authentication,

event creation, event browsing, RSVP submission, and logout without affecting existing system functionality.

7.3 User Interface and Navigation Validation

The screenshot displays the 'My RSVPs' page on the Campus Event Board. The page features a dark blue header with the 'Campus Event Board' logo and navigation links for 'Events' and 'My RSVPs'. A user profile for 'Jane Student' with a 'Logout' button is in the top right. Below the header, the 'STUDENT DASHBOARD' section shows the title 'My RSVPs' and a 'Browse events' button. A table lists the user's RSVPs:

EVENT	DATE	VENUE	STATUS	ACTION
Wellness & Mental Health Awareness Day Interactive sessions on mental health, stress management, healthy living, and student wellbeing. <i>Health</i>	Sep 29, 2026 10:00 AM	Student Wellness Centre	Going	Cancel RSVP
Cultural Heritage Festival Celebrate cultural diversity through traditional music, dance, food exhibitions, and storytelling. <i>Culture</i>	Oct 9, 2026 2:00 PM	Student Centre Grounds	Going	Cancel RSVP
Talent & Music Showcase Students showcase singing, dancing, comedy, spoken word, and other talents in a lively evening ev... <i>Entertainment</i>	Oct 15, 2026 6:00 PM	University Amphitheatre	Going	Cancel RSVP

Figure 34: Navigation validation confirming successful page transitions throughout the Campus Event Board.

7.4 Final System Performance Validation

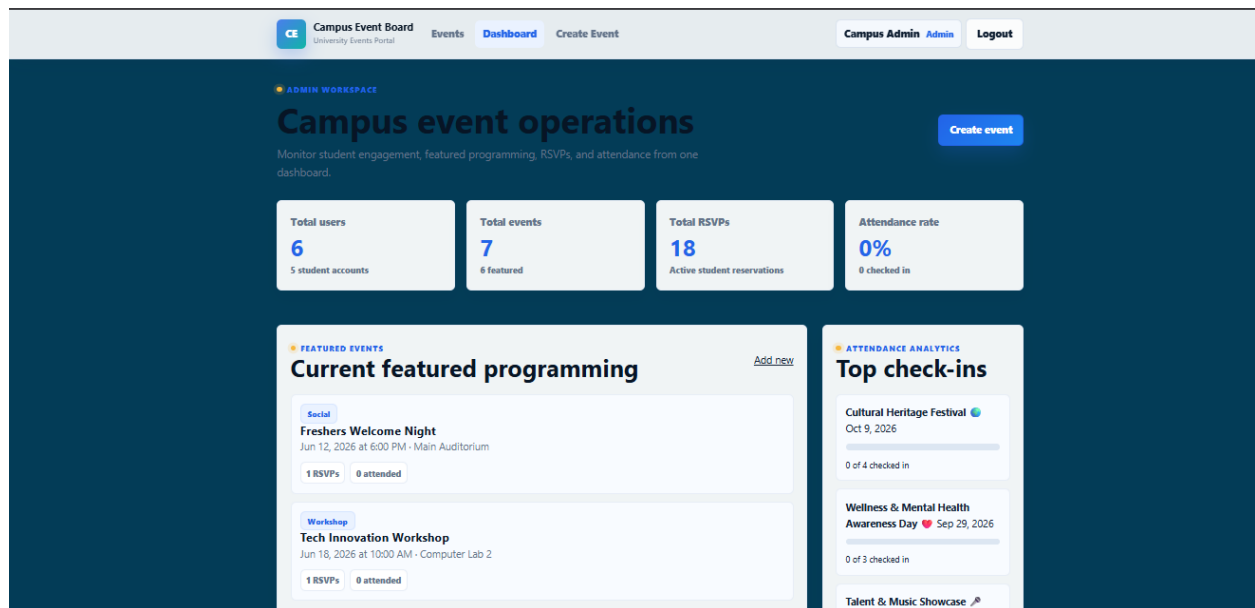


Figure 35: Final system performance validation using the administrator dashboard.

7.5 Final Project Verification

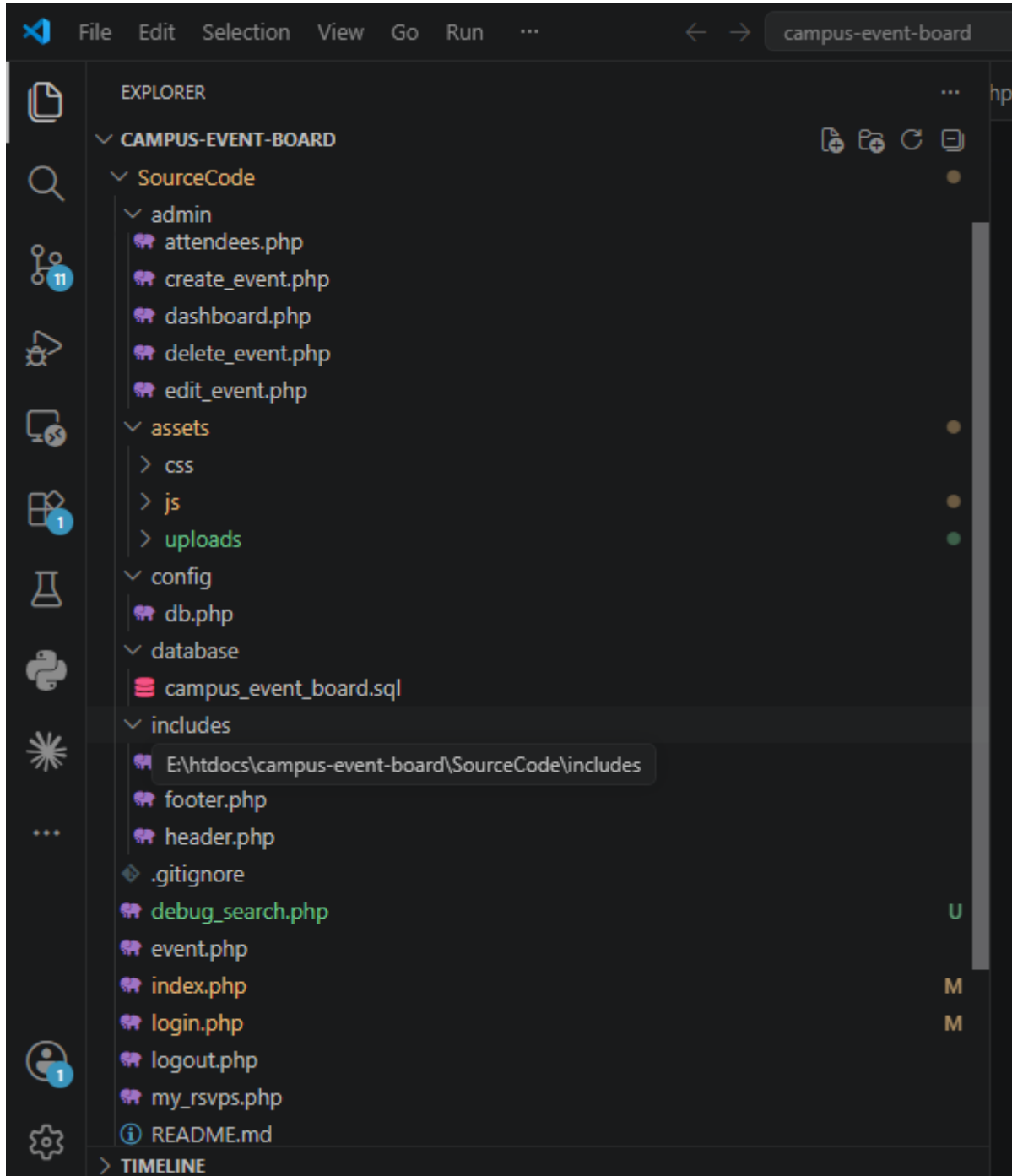


Figure 36: Final Campus Event Board project structure before submission.

7.6 Creative Enhancement

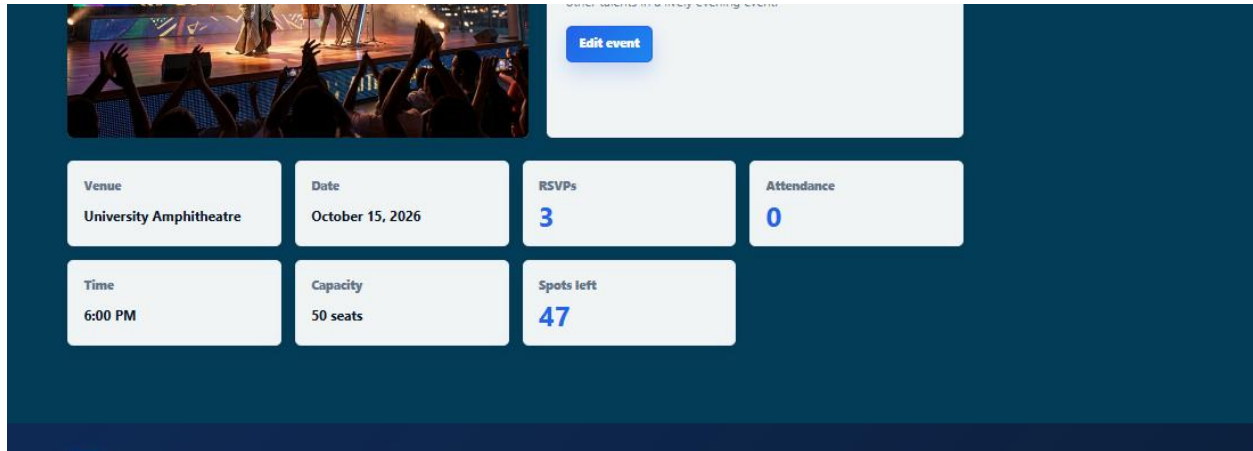


Figure 37: Dynamic event capacity indicator displaying remaining available spaces.

7.6.1 Reflection

During Week 7, I completed final system testing, validated responsive design, improved user navigation, and verified that the Campus Event Board functioned correctly across different pages. These activities strengthened my understanding of testing, usability, and preparing a web application for deployment.

8 WEEK 8: Responsive Web Design and Mobile-First Development

8.1 Mobile-First CSS Optimization Using Media Queries

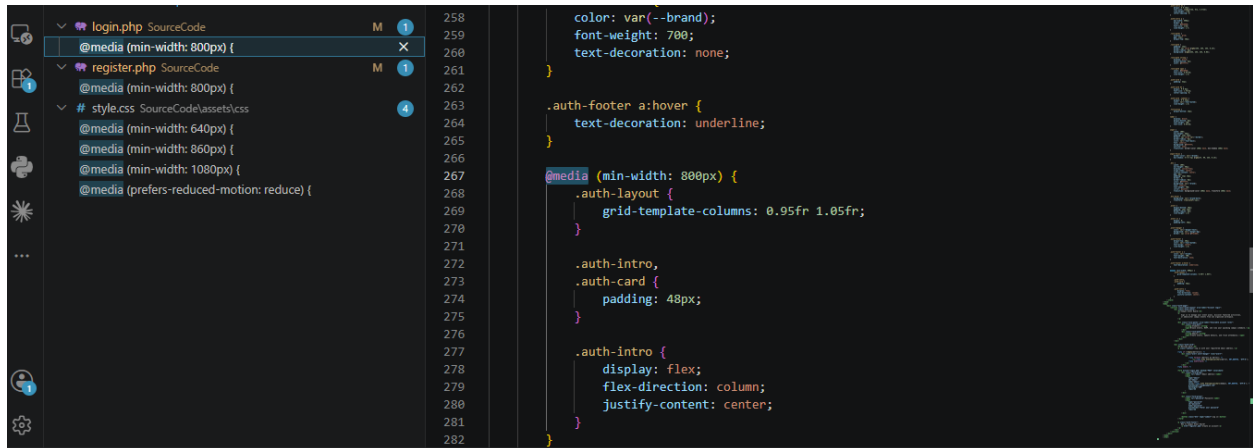
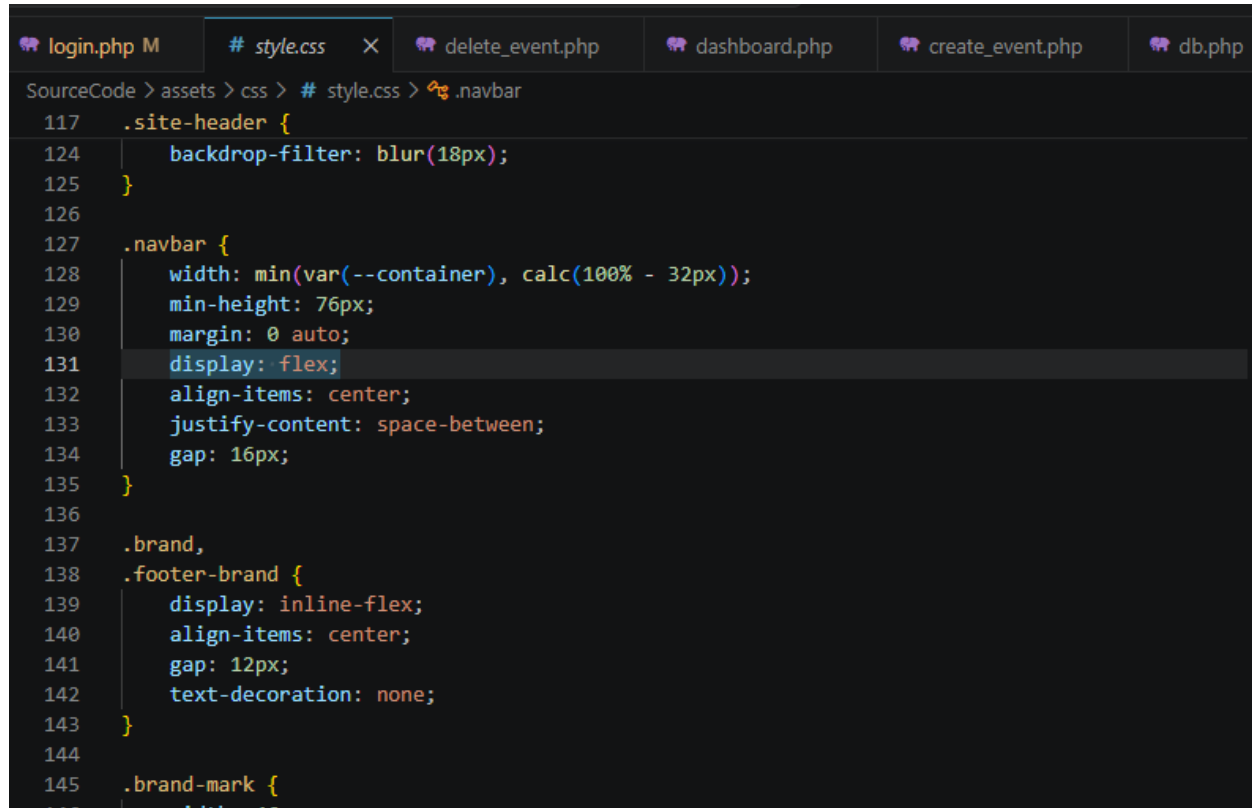


Figure 38: CSS media queries used to optimize the Campus Event Board for responsive mobile-first layouts.

CSS media queries were implemented to adapt the layout of the Campus Event Board to different screen sizes. The responsive rules adjust interface elements for mobile, tablet, and desktop devices, improving accessibility and user experience without changing the application's functionality.

8.2 Responsive Layout Using Flexbox and CSS Grid



```
login.php M # style.css X delete_event.php dashboard.php create_event.php db.php
SourceCode > assets > css > # style.css > .navbar
117 .site-header {
124     backdrop-filter: blur(18px);
125 }
126
127 .navbar {
128     width: min(var(--container), calc(100% - 32px));
129     min-height: 76px;
130     margin: 0 auto;
131     display: flex;
132     align-items: center;
133     justify-content: space-between;
134     gap: 16px;
135 }
136
137 .brand,
138 .footer-brand {
139     display: inline-flex;
140     align-items: center;
141     gap: 12px;
142     text-decoration: none;
143 }
144
145 .brand-mark {
146     width: 46px;
```

Figure 39: Flexbox and CSS Grid implementation for creating responsive layouts within the Campus Event Board.

8.2.1 Reflection

During Week 8, I improved the responsiveness of the Campus Event Board by reviewing responsive CSS techniques and optimizing layouts for different screen sizes. This enhanced my understanding of mobile-first web development and the importance of creating consistent user experiences across multiple devices.